Resources

MATH 107: Introduction to Linear Algebra

Daily reading assignments are listed with each meeting on Lectures.

Textbook

The main text is Stephen Boyd and Lieven Vandenberghe, *Introduction to Applied Linear Algebra: Vectors, Matrices, and Least Squares* (VMLS). The book is free to read and download from the authors.

You do **not** need to buy Gilbert Strang, *Introduction to Linear Algebra*. A few meetings use three freely posted MIT files. The ILA, 6th edition resource page distinguishes those sample sections from the commercial book.

When	File	What to read
Nov 4	§3.5 Dimensions of the Four Subspaces	Rank, column space, null space, basis, and dimension. Not the whole section: row space and left nullspace are enrichment, not quiz material.
Nov 16–20	§6.1 Introduction to Eigenvalues	$Ax = \lambda x$, dynamics, the characteristic equation, and the 90° rotation with eigenvalues i and $-i$. This PDF is only §6.1; later Chapter 6 sections listed at the start of the file are not free from this folder.
Dec 4	SVD slides	Selected slides (not all 17 pages) on SVD, low-rank approximation, data compression, least squares, and image processing.

Software

We use **Octave**. Each lab will have an Octave version and a Python/Colab version; pick one language and stay with it this term. Lab 0 assumes no prior coding. Bring a laptop to lab days, or let me know at least a day ahead to bring a campus one for you.

Primary Octave reference: Jason Lachniet, *Introduction to GNU Octave: A Brief Tutorial for Linear Algebra and Calculus Students* (3rd ed., 2020). Free PDF, licensed CC BY-SA 4.0. Author, title, and PDF access are also listed at the Open Textbook Library.

Use Lachniet for syntax, commands, and selected exercises. The **lab handouts** are the assigned work: they tell the application story (similarity, transformations, dynamics, eigenbehavior, least squares, and the three-frame video capstone). Daily chapter assignments are on Lectures; it is all one PDF, not a new link each lab.

Lab	Lachniet chapters
Lab 0	Ch. 1 Getting Started — interface, variables, vectors/matrices, basic operations, plotting
Lab 1	Ch. 1 — vectors, matrix storage, indexing, plots (plus our distance/nearest-neighbor scaffold)
Lab 2	Ch. 2 — matrices and linear-algebra computations (plus our transformation plotting code)
Systems / RREF (Oct 14–16)	Ch. 2 — solving systems, Gaussian elimination/RREF, linear-system commands
Lab 3	Ch. 2 matrix operations, plus loops/programming from the later introductory programming portions
Lab 4	Ch. 5 — eigenvalues/eigenvectors, Markov chains, diagonalization
Lab 5	Ch. 2, especially least-squares / data-fitting exercises
Capstone	Ch. 5 SVD, plus Ch. 7 project on SVD / image compression

Optional reference: the GNU Octave documentation (not assigned reading). Download the program from octave.org.

Python / Google Colab

If you choose Python, use Google Colab for all Python labs. Colab is a hosted Jupyter notebook; you do not need to install Python locally. Each lab will have a Colab version alongside the Octave version.

Role	Tool	Link
Where we write and run Python	Google Colab	colab.google
Organize and prepare tabular data	Polars	Getting started
Vectors, matrices, and linear algebra	NumPy	Beginner guide; later, linear algebra
Visualize and communicate results	Plotnine	Introductory guide

Polars is for real tables: reading files, selecting columns, filtering, transforming, and handing numerical columns to NumPy. **NumPy** is for arrays, indexing, and later `solve`, `eig`, `lstsq`, `svd`, norms, and determinants. **Plotnine** is for scatterplots, line plots, fitted relationships, residual plots, clustering graphics, and other standard figures; it uses a grammar-of-graphics syntax and works directly with Polars DataFrames.

As with Lachniet, these pages support syntax. The lab handouts are the assigned work.

Campus

- Cal Poly Humboldt
- Library
- Student Disability Resource Center
- Counseling and Psychological Services

Tutoring and drop-in help: check the current campus learning-center listings once the term site is posted, or ask in office hours.